# GAUTHAM KRISHNA GUDUR

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## Research Interests

I am interested in building data-efficient, data-centric, and human-centered machine learning systems.

- **Multimodal and multisensory AI:** human signals (temporal, physiological, wearable, acoustic) for health.
- **Cross-modal representation learning:** modality gaps and alignment across time-series, language, and vision.
- **Adaptive and agentic AI:** continual learning, personalization, and few-shot adaptation; self-evolving agents.
- **LLMs and specialized foundation models:** efficient pre- and post-training and inference across modalities.
- **Efficient learning and reasoning:** data-, sample-, parameter-, and token-efficient methods, including PEFT.
- **Data-centric learning:** informative sample selection, data curation, and uncertainty quantification.

## Education

**Ph.D. in Electrical and Computer Engineering** — Aug. 2023 – Present
The University of Texas at Austin — Austin, TX, USA
Advisor: Prof. Joydeep Ghosh | GPA: 3.9/4.0

**B.Tech. in Information Technology** — June 2017
Anna University — Chennai, India
Advisors: Prof. Srinivasan R, Prof. Bhalaji N

## Selected Publications

* equal contribution and joint lead authorship. Google Scholar | **416** citations | h-index **10** | i10-index **10**

### *Conference, Journal, and Workshop Papers*

[1] **Latent-LoRA: Compact Latent-Space Adapters with Gradient-Free Routing for Continual Learning**
R.R. Azghan, **Gautham Krishna Gudur**, P. Turaga, G. Pedrielli, and H. Ghasemzadeh
*EMNLP Findings* 2026

[2] **RAST: Resolution-Aware Privileged Structure Transfer for Low-Resolution Audio Activity Recognition**
J.H. Park, **Gautham Krishna Gudur**, Y. Shen, D. Liang, and E. Thomaz
*ACM MobiCom Workshop on Human-Centered Sensing, Modeling, and Intelligent Systems (HumanSys)* 2026

[3] **WellComp 2026: Ninth International Workshop on Computing and Software Systems for Well-Being**
**Gautham Krishna Gudur**, H. Yoon, Y.Y. Wu, T. Xia, and H. Jia
*ACM UbiComp Companion* 2026

[4] **PCL: Partitioned Continual Learning via Unsupervised Latent Experts for Audio Classification**
**Gautham Krishna Gudur***, M. Malu*, T. Khandait, R.R. Azghan, A. Rayas, P. Turaga, J. Ghosh, et al.
*ICML Workshop on Machine Learning for Audio* 2026

[5] **Gated Adaptation for Continual Learning in Human Activity Recognition**
R.R. Azghan, **Gautham Krishna Gudur**, M. Malu, E. Thomaz, G. Pedrielli, P. Turaga, and H. Ghasemzadeh
*IEEE Internet of Things Journal* 2026

[6] **AttenGluco: Multimodal Transformer-Based Blood Glucose Forecasting on AI-READI Dataset**
E. Farahmand, R.R. Azghan, N.T. Chatrudi, E. Kim, **Gautham Krishna Gudur**, E. Thomaz, G. Pedrielli, P. Turaga, and H. Ghasemzadeh
*International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)* 2025

[7] **SVFT: Parameter-Efficient Fine-Tuning with Singular Vectors**
V. Lingam*, A. Tejaswi*, A. Vavre*, A. Shetty*, **Gautham Krishna Gudur***, J. Ghosh, A. Dimakis, E. Choi, A. Bojchevski, and S. Sanghavi
*Neural Information Processing Systems (NeurIPS)* 2024
*NeurIPS Workshop on Fine-Tuning in Modern Machine Learning: Principles and Scalability (FITML)*
*ICML Workshop on Efficient Systems for Foundation Models (ES-FoMo)*
*ICML Workshop on Advancing Neural Network Training (WANT)* [ Oral]

[8] **Dataset Distillation for Audio Classification: A Data-Efficient Alternative to Active Learning**
**Gautham Krishna Gudur** and E. Thomaz
*NeurIPS Efficient Natural Language and Speech Processing Workshop (ENLSP)* 2024

[9] **Can Calibration Improve Sample Prioritization?**
G. Tata*, **Gautham Krishna Gudur***, G. Chennupati, and M.E. Khan
*NeurIPS Workshop on Human in the Loop Learning (HILL)*
*NeurIPS Workshop on Has It Trained Yet? (HITY)* 2022

[10] **Data-Efficient Automatic Model Selection in Unsupervised Anomaly Detection**
**Gautham Krishna Gudur**, R. Raaghul, K. Adithya, and S. Vasudevan
*IEEE International Conference on Machine Learning and Applications (ICMLA)* 2022

[11] **Zero-Shot Federated Learning with New Classes for Audio**
**Gautham Krishna Gudur** and S.K. Perepu
*INTERSPEECH* 2021
*ICLR Workshop on Distributed and Private Machine Learning (DPML)*
*ICLR Workshop on Hardware Aware Efficient Training (HAET)*

[12] **Bayesian Active Learning for Wearable Stress and Affect Detection**
A. Ragav* and **Gautham Krishna Gudur***
*NeurIPS Workshop on Machine Learning for Mobile Health (MLMH)* 2020

[13] **Resource-Constrained Federated Learning with Heterogeneous Labels and Models for Human Activity Recognition**
**Gautham Krishna Gudur** and S.K. Perepu
*IJCAI Workshop on Deep Learning for Human Activity Recognition (DL-HAR)*
*ACM KDD Workshop on Artificial Intelligence of Things (AIoT)* 2020

[14] **Federated Learning with Heterogeneous Labels and Models for Mobile Activity Monitoring**
**Gautham Krishna Gudur** and S.K. Perepu
*NeurIPS Workshop on Machine Learning for Mobile Health (MLMH)* 2020

[15] **A Dynamically Adaptive Movie Occupancy Forecasting System with Feature Optimization**
S. Venkataramani, A. Ramesh, S. Sharan Sundar, A.K. Jain, **Gautham Krishna Gudur**, and V. Vijayaraghavan
*IEEE ICDM Workshop on Learning and Mining with Industrial Data (LMID)* 2019

[16] **A Vision-based Deep On-Device Intelligent Bus Stop Recognition System**
**Gautham Krishna Gudur**, A. Ramesh, and R. Srinivasan
*ACM UbiComp Workshop on Pervasive Urban Applications (PURBA)* 2019

[17] **Label Frequency Transformation for Multi-Label Multi-Class Text Classification**
A.K. Raghavan, V. Umaashankar, and **Gautham Krishna Gudur**
*KONVENS GermEval Shared Task 1, Subtask (a)* [🏆 Winner, post-evaluation phase] 2019

[18] **ActiveHARNet: Towards On-Device Deep Bayesian Active Learning for Human Activity Recognition**
**Gautham Krishna Gudur**, P. Sundaramoorthy, and V. Umaashankar
*ACM MobiSys Workshop on Embedded and Mobile Deep Learning (EMDL)* 2019

[19] **HARNet: Towards On-Device Incremental Learning using Deep Ensembles on Constrained Devices**
P. Sundaramoorthy, **Gautham Krishna Gudur**, M.R. Moorthy, R.N. Bhandari, and V. Vijayaraghavan
*ACM MobiSys Workshop on Embedded and Mobile Deep Learning (EMDL)* 2018

### *Preprints*

[20] **Counterfactual Self-Evolving Agents for Evidence-Grounded Reasoning**
X. Han, Y. Wang, C. Chen, W. Dai, **Gautham Krishna Gudur**, S. Li, H.H. Chung, G.D. Hager, J. Ghosh, P. Liang, and S. Saria
*In submission to ICLR 2027* 2026

[21] **SensorHarness: Optimizing How LLMs Sense the World**
H. Yoon, S. Im, P. Langer, T. Lee, **Gautham Krishna Gudur**, J. Kim, X. Han, Y.Y. Wu, N. Zumarraga, Y. Zhang, R. Müller, J. Ghosh, Y. Yang, and S.J. Lee
*To be submitted* 2026

[22] **CLAD-Net: Continual Activity Recognition in Multi-Sensor Wearable Systems**
R.R. Azghan, **Gautham Krishna Gudur**, M. Malu, E. Thomaz, G. Pedrielli, P. Turaga, and H. Ghasemzadeh
*In submission to ACM Transactions on Intelligent Systems and Technology* 2026

[23] **GluMind: Multimodal Parallel Attention and Knowledge Retention for Robust Cross-Population Blood Glucose Forecasting Systems**
E. Farahmand, R.R. Azghan, N.T. Chatrudi, V.Y.A. Baidoo, E. Kim, **Gautham Krishna Gudur**, et al.
*In submission to ACM Transactions on Computing for Healthcare* 2026

### *Patents*

[1] **Efficient Backpropagation Free Test-Time Adaptation on Foundation Models for PPG**
**Gautham Krishna Gudur**, S. Chatterjee, and M. Malekzadeh
*Filed*

[2] **Method and Apparatus for Approach Recommendation with Threshold Optimization in Unsupervised Anomaly Detection**
**Gautham Krishna Gudur**, K. Adithya, R. Raaghul, and S. Vasudevan
*WO2023166515A1* 2023

[3] **Distributed Machine Learning with New Labels using Heterogeneous Label Distribution**
**Gautham Krishna Gudur** and S.K. Perepu
*WO2022162677A1* 2022

[4] **Federated Learning using Heterogeneous Labels**
S.K. Perepu and **Gautham Krishna Gudur**
*WO2022013879A1* 2022

## Research and Work Experience

**The University of Texas at Austin** – Graduate Research Assistant — Austin, TX
Advisors: Prof. Joydeep Ghosh, Prof. Edison Thomaz — Aug. 2023 – Present

- Built benchmarks and multi-agent systems for context-aware multimodal health personalization at test time.
- Proposed SVFT (Singular Vectors guided Fine-Tuning), a Pareto-optimal PEFT method [with Prof. Sanghavi].
- Designed continual learning via unsupervised latent partitioning for audio sensing, LoRA adapters, multi-sensor HAR, and cross-population glucose prediction; reduced labeling costs up to 3000× with dataset distillation.
- Developed multimodal and cross-modal alignment techniques for time-series, language, health, and audio.

**Bosch Research** – Foundation Models Research Intern (BCAI) — Sunnyvale, CA
Mentors: Gorden Li, Wenbin He, Liu Ren — May 2026 – Aug. 2026

- Equipped multimodal language models with general-purpose time-series understanding via tool-calling agents.

**Nokia Bell Labs** – Research Intern (Device Intelligence team) — Cambridge, UK
Mentors: Soumyajit Chatterjee, Mohammad Malekzadeh, Fahim Kawsar — May 2025 – Aug. 2025

- Improved generalizability of PPG-based health foundation models using test-time adaptation.

**Independent Research** — Dec. 2018 – May 2023

- Analyzed how calibration affects sample prioritization during training [with Prof. Emtiyaz Khan, RIKEN].
- Worked on Bayesian active learning for on-device human activity recognition (HAR) and wearable sensing.

**Ericsson R&D, Global AI Accelerator (GAIA)** – Data Scientist III — Chennai, India
Mentors: Shrihari Vasudevan, M J Prasath — Feb. 2019 – Apr. 2023

- Worked on ML for network intelligence and telecom, leading to publications, patents, and AI-driven solutions.
- Advanced Ericsson's AI-Native telecom stack: 3GPP standardization for federated learning and multi-vendor model sharing, 5G NWDAF mobility and connectivity modeling, and Bayesian anomaly detection (E-ADF).
- Built telco-AI platforms: the Ericsson LLM Hub with domain-rich data and LLMs, language translation, active learning for customer-symptom modeling, and object detection for cell-site infrastructure failures.
- Designed zero-shot federated learning frameworks to handle new heterogeneous classes and models.

**SmartCardia (EPFL)** – Machine Learning Engineer — Chennai, India
Mentor: Srinivasan Murali — May 2018 – Nov. 2018

- Implemented gradient-boosted ensembles and LSTMs for regression and classification on imbalanced wearable clinical time-series; engineered biomarker features for sleep apnea, troponin, blood pressure, and hemoglobin.

**Solarillion Foundation** – Research Assistant — Chennai, India
Mentor: Vineeth Vijayaraghavan — Feb. 2016 – May 2018

- Led development of HARNet – deep ensembles for activity recognition with on-device incremental learning.
- Designed dynamic gesture recognition models with efficient feature engineering on a $5 Raspberry Pi Zero.

**SSN College of Engineering** – Undergraduate Student Researcher — Chennai, India
Mentors: Prof. Srinivasan R, Prof. Bhalaji N — Feb. 2015 – Mar. 2017

- Developed a CNN-based bus stop recognition system with ∼96% accuracy in dynamic Indian environments.
- Led an HCI project on Neurocinematics to classify real-time cognitive responses of film viewers via EEG.

## Honors, Awards, and Fellowships

- **Oral presentation** – SVFT at the ICML WANT Workshop 2024.
- **Graduate Ph.D. Fellowship** – Cockrell School of Engineering, UT Austin.
- **Winner, GermEval Shared Task 1 (Subtask a)** – KONVENS 2019, post-evaluation phase.
- **Ericsson Top Performance Competition 2020** – Operational Excellence category, for the AIB project.
- **Full registration grants** – ICLR 2021, NeurIPS 2020, and OxML 2020.
- **Top 1 percentile** – HackerRank Problem Solving (Advanced), Algorithms domain.
- **Undergraduate research grant** – INR 25,000 from SSN College of Engineering.

## Talks and Presentations

- Machine Learning and Ubiquitous Computing
  *SSN College of Engineering* June 2022
- Heterogeneous Zero-Shot Federated Learning with New Classes for On-Device Audio Classification
  *MobiUK* July 2021
- Zero-Shot Federated Learning with New Classes for Audio Classification
  *Eastern European Machine Learning Summer School (EEML)* July 2021
- Telecom-Specific Language Translation using GCP
  *Ericsson/Google Cloud Day* May 2021
- Resource-Constrained Machine Learning for Ubiquitous Computing Applications
  *Flipped by GAIUS* Sept. 2020
- ActiveHARNet: Towards On-Device Deep Bayesian Active Learning for Human Activity Recognition
  *Eastern European Machine Learning Summer School (EEML)* July 2020

## Teaching and Mentoring

### *Teaching*

**The University of Texas at Austin** – Graduate Teaching Assistant 2025 – Present

- ECE 381K Applied Machine Learning, taught by Prof. Joydeep Ghosh (∼40 students) Fall 2026
- ECE 381V Deep Learning, taught by Prof. Joydeep Ghosh (∼50 students) Fall 2025
- ECE 382V Human Signals: Sensing and Analytics, taught by Prof. Edison Thomaz (∼80 students) Spring 2025
- Designed assignments, exams, and quizzes; mentored students on paper reviews, presentations, and projects.

**Solarillion Foundation** – Teaching Assistant Feb. 2017 – May 2018

- Mentored 10+ students on embedded ML research projects and assignments.

### *Mentoring*

Aadharsh Aadithya (UT Austin) 2026 – Present
*Topic: Discrete diffusion models for time-series data*

Ji Hwan Park (UT Austin) 2025 – 2026
*Topic: Audio sensing and multimodal alignment*

Abhijith Ragav (Georgia Tech → Amazon) 2020
*Topic: Bayesian active learning for wearable stress and affect detection*

Prahalathan Sundaramoorthy (USC → HP Labs) 2018
*Topic: Bayesian active learning for activity recognition*

Nidhi Bhandari (UCLA) 2017 – 2018
*Topic: Deep learning and human activity recognition*
Manav Rajiv Moorthy (University of Washington → Amazon) 2017 – 2018
*Topic: Deep learning and human activity recognition*

## Professional Service

### *Workshop Organizer*

- WellComp 2026 workshop at ACM UbiComp/ISWC 2026

### *Program Committee / Reviewer*

- Pervasive and Mobile Computing (Journal) 2026
- International Conference on Acoustics, Speech and Signal Processing (ICASSP) 2025 – 2026
- INTERSPEECH 2026
- IEEE International Workshop on Machine Learning for Signal Processing (MLSP) 2026
- ACM International Joint Conference on Pervasive and Ubiquitous Computing (UbiComp/ISWC) 2025
- ACM International Conference on Multimodal Interaction (ICMI) 2025
- ACM UbiComp Workshop on Computing and Software Systems for Well-Being (WellComp) 2026
- ICML Workshop on Efficient Systems for Foundation Models (ES-FoMo) 2024
- ICLR Workshop on Distributed and Private Machine Learning (DPML) 2021
- NeurIPS Workshop on Machine Learning for Health (ML4H) 2019 – 2020
- KONVENS GermEval 2019

### *Other Service*

- Event Organizer, Data Nuggets – a data science contest at Invente, SSN College of Engineering 2016

## Posters and Extended Abstracts

[1] Uncertainty-Aware Chain-of-Thought Reasoning in Large Language Models, *AI in Health*, UT Austin.

[2] Adaptive Federated Learning in Conceptually Drifting Environments, *Applied ML*, UT Austin.

[3] Heterogeneous Zero-Shot Federated Learning with New Classes for On-Device Audio Classification, MobiUK 2021.

[4] Bayesian Active Learning for Wearable and Mobile Health, NeurIPS Europe BDL Meetup 2020.

[5] Handling Real-time Unlabeled Data in Activity Recognition using Deep Bayesian Active Learning and Data Programming, MobiUK 2019, University of Oxford.

[6] Neurocinematics: The Intelligent Review System, CBC 2015, IIT Gandhinagar.

## Summer Schools

5th Summer School on Artificial Intelligence, CVIT, IIIT Hyderabad Aug. 2022
– Computer vision and machine learning.
Eastern European Machine Learning Summer School (EEML) July 2021
– Poster on zero-shot federated learning with new classes; unconference talk on continual learning.
Oxford Machine Learning Summer School (OxML) Aug. 2020
– Deep learning and healthcare; attended with a full-fee waiver.
Eastern European Machine Learning Summer School (EEML) July 2020
– Poster on ActiveHARNet.

## Technical Skills

**Languages:** Python, C++, HTML/CSS, Bash, SQL, JavaScript
**Tools:** Git, LaTeX, Claude Code, Docker, GCP
**Libraries:** PyTorch, TensorFlow, Hugging Face Transformers, Keras, NumPy, Pandas, Scikit-learn, PySpark

## Technical Coursework and MOOCs

### *Graduate Coursework*

Generative Models in Machine Learning | Advanced Computer Vision | Statistical Methods I & II | Ethics of AI
Applied Machine Learning | Fine-Tuning LLMs | AI in Health | Spoken Language Technologies | Human Signals

### *MOOCs and Certifications*

Machine Learning Specialization, University of Washington (Coursera)
Foundations | Regression | Classification | Clustering & Retrieval
Machine Learning, Stanford University (Coursera)
Algorithmic Toolbox and Data Structures, UC San Diego (Coursera)
R Programming, Johns Hopkins University (Coursera)
Problem Solving – Advanced, Intermediate, Basic (HackerRank)